Jason Hkayem

81-654045 | jasonhkayem1@gmail.com | LinkedIn | Portfolio

CS graduate (June 2026) with hands-on experience as a machine learning engineer, building and evaluating embedding, deep learning, and graph-based recommendation systems in production. Also comfortable across the stack, from cloud infrastructure to backend APIs. Open to technical and hybrid roles in ML, AI, operations, or product.

Education

Bachelor of Computer Science, Holy Spirit University of Kaslik | 2023–2026 | GPA: 3.67, Honors | **Baccalaureate in SG,** College Saint-Joseph des Filles de la Charite | 2006–2021

Skills

Languages: Arabic, English, French | **Programming:** Python, Java, C++, C, JavaScript, HTML, CSS, PHP
Frameworks: Flask, Streamlit, PyTorch, HuggingFace, SentenceTransformers, FAISS, NumPy, Pandas, scikit-learn | **Tools:** AWS, Docker, PostgreSQL, MLFlow, Airflow, n8n, Power BI

Work Experience

Machine Learning Engineer Intern | Anghami / OSN+ June 2026 – Present

- Built and evaluated lyrics-embedding pipelines across genre/language classification, a third-party benchmark, and curated mood playlists to test whether lyrical signal could strengthen collaborative-filtering recommendations
- Designed and trained PyTorch neural networks, including multi-task learning architectures, on lyrics embeddings to predict mood/playlist fit
- Prototyped and scaled a Graph Neural Network-based collaborative-filtering model as an alternative recommendation approach, training on the full production dataset
- Evaluated optimization, systems, and architecture tradeoffs (training cost vs. accuracy, model choice, pipeline design) throughout each project to keep experimental work grounded in production feasibility

Cloud AI/ML Engineer Intern | Zero&One Summer 2025

- Designed and exposed RESTful CRUD APIs using Flask within a modular, service-oriented backend architecture
- Integrated AWS Bedrock into a serverless application to enable generative AI capabilities within a scalable cloud infrastructure
- Built document embedding and retrieval pipelines using RAG architecture to enable semantic search over domain-specific content

Waiter | Anthony's Foodhub May 2021–Present

- Demonstrated reliability and time management while balancing work with full-time studies in a fast-paced, team-oriented environment

Projects

AI Triage Workflow | *n8n, LLM Orchestration*

- Designed a deterministic four-stage LLM pipeline (ingestion → parallel classification/enrichment → parsing → decision engine) to transform unstructured support messages into structured, machine-readable records for downstream automation
- Implemented parallel LLM calls with strict JSON schemas to classify messages by category, priority, and confidence while extracting structured fields with explicit null-handling to eliminate parsing failures
- Built rule-based decision engine consolidating routing logic, priority computation, and escalation rules into a single, centralized, auditable code node for consistency and maintainability

CloudQ | *RAG-based learning platform chatbot*

- Developed a RAG-based chatbot using AWS services to answer course-specific queries against embedded document knowledge bases
- Architected a modular retrieval layer decoupled from the generation layer, enabling swapping of embedding models or LLM providers without restructuring the pipeline

Stride | *AI running platform | Flask, PostgreSQL, OpenAI, Strava OAuth2*

- Engineered an AI coaching layer using OpenAI's API to dynamically generate personalized weekly training plans calibrated to individual user goals and activity history
- Implemented Strava OAuth2 to sync real athlete data (lap splits, pace, heart rate) into a unified schema; built full JWT auth system with bcrypt hashing and secure forgot-password email flow
- Built a social layer (feeds, likes, comments, follows, clubs, audit logging) and visualized activity data with Chart.js and Leaflet.js GPS maps; deployed as a production SPA with vanilla JS hash-router

Volunteering

- **Scouts du Liban** | Member (2015–Present), Patrol Chief (2019–2021), Team Chief (2024–Present)